

RESULTS OF HDM DECOMPOSITION FOR 18 OCTANE ISOMERS

Program parameters: relative error $\varepsilon = 9.99 \times 10^{-17}$, maximum index $Q_{\max} = 120$, experimental uncertainty $\Delta S = 0.10 \text{ cal mol}^{-1} \text{ K}^{-1}$.

For each isomer, three MHD integer sequences (coefficients $m_1, m_2, m_3, \dots$ up to the last non-zero term) are given, obtained from the entropy interval $S \pm 0.10 \text{ cal mol}^{-1} \text{ K}^{-1}$: lower ($S - 0.10$), central (S), and upper ($S + 0.10$).

1. *n*-Octane

lower: 0, 0, 0, 1, -1, -1, 0, 1, -1, -1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, -1, -1, -1, -1, 0, -1, -1, 0, 1, 0, 1, 1, -1, -1, -1, 1, 0, 1
central: 0, 0, 0, 1, -1, -1, 0, 1, -1, -1, -1, 0, 0, -1, 0, 1, -1, 0, -1, 0, 1, 0, 0, 1, -1, -1, -1, 1, 0, 1, -1, -1
upper: 0, 0, 0, 1, -1, -1, 0, 0, 1, 1, 1, -1, 1, 1, -1, 0, 0, 1, 1, -1, 1, 1, -1, -1, 1, 0, 0, -1, 0, 0, 0, 1, 1, -1, 1, 0, 1

2. 2-Methylheptane

lower: 0, 0, 0, 1, -1, -1, 1, -2, 2, -2, 1, 1, 1, 1, 1, 1, -1, -1, 0, 1, -1, 0, -1, -1, 0, 0, 1, -1, 1, 1, 1, 1, 0, 1, -1, -1, 0, 1
central: 0, 0, 0, 1, -1, -1, 0, 1, 1, 1, 1, 0, 1, 1, 1, -1, 1, -1, 0, 1, -1, 1, 1, -1, -1, -1, 0, 1, 1, 1, 1, -1, -1, 1, 1, 1, -1
upper: 0, 0, 0, 1, -1, -1, 0, 1, 1, 1, 0, -1, 0, 1, -1, 1, 0, -1, 0, 0, -1, 1, -1, -1, -1, 1, 0, 0, 1, 0, 1, 1, 0, 1, 0, 1, -1

3. 3-Methylheptane

lower: 0, 0, 0, 1, -1, -1, 0, 1, -1, 1, 0, 1, -1, -1, 1, 1, 0, 0, 0, -1, -1, -1, -1, 0, 1, -1, -1, -1, -1, 1, 0, 1, -1, -1, 0, -1
central: 0, 0, 0, 1, -1, -1, 0, 1, -1, 1, -1, -1, 1, 0, 0, -1, 0, 1, 0, 0, 0, 1, -1, 1, 1, -1, 0, 0, 1, -1, 1, 1, 0, -1, -1, 1, 1
upper: 0, 0, 0, 1, -1, -1, 0, 1, -1, 0, 0, 1, 0, 1, -1, 0, 0, 0, 1, -1, 1, 1, -1, 1, -1, 0, 0, 0, -1, 1, 0, -1, -1, -1, -1, 1, -1

4. 4-Methylheptane

lower: 0, 0, 0, 1, -1, -1, 1, -1, -1, 1, 0, 0, -1, 1, 0, 0, -1, 0, -1, 0, -1, -1, -1, 0, 0, 1, 1, 0, 1, 0, 0, 0, -1, -1, 0, 1
central: 0, 0, 0, 1, -1, -1, 1, -1, -1, 0, 1, 1, 1, 0, 1, 0, -1, -1, -1, 1, 0, 1, -1, 0, -1, -1, -1, 0, 1, -1, 1, 1, -1, -1, 0, 0, 1, -1
upper: 0, 0, 0, 1, -1, -1, 1, -1, -1, 0, 0, 0, 0, -1, 1, 1, -1, 1, 0, 1, 0, 1, 1, 0, -1, -1, 0, 0, 0, 1, -1, 1, 0, 0, -1, -1

5. 2,2-Dimethylhexane

lower: 0, 0, 0, 1, -1, 0, -1, -1, 0, -1, 0, 0, -1, 1, 0, 1, 0, 0, -1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, -1, 1
central: 0, 0, 0, 1, -1, 0, -1, -1, -1, 1, 1, 1, 0, 0, -1, 1, 1, -1, -1, -1, 1, 0, 1, 1, 1, -1, 1, 1, -1, -1, 0, -1, 0, -1, 1, 0, -1
upper: 0, 0, 0, 1, -1, 0, -1, -1, -1, 1, 0, -1, 0, 0, 1, -1, 1, 0, 1, -1, 1, 0, 0, 1, -1, -1, -1, 0, -1, 0, 1, 0, 0, 0, -1, 1

6. 2,3-Dimethylhexane

lower: 0, 0, 0, 1, -1, -1, 1, -1, -1, 1, 1, 1, 1, -1, 1, 0, 1, 1, 1, 1, 0, 1, 0, 1, -1, -1, 0, -1, -1, 0, -1, -1, 1, 0, 1, 0, 0, 1
central: 0, 0, 0, 1, -1, -1, 1, -1, -1, 1, 1, 0, -1, 1, -1, 1, -1, 1, 1, 0, -1, 0, 0, 0, 0, 0, 0, 0, 0, -1, -1, 0, 0, 1, -1, -1
upper: 0, 0, 0, 1, -1, -1, 1, -1, -1, 1, 0, 1, 1, -1, 1, -1, 0, 1, -1, 0, 1, 0, 0, 1, 0, -1, -1, 1, -1, 1, 1, 0, 0, -1, 1, -1, 1, 1, 1

7. 2,4-Dimethylhexane

lower: 0, 0, 0, 1, -1, -1, 1, 0, 0, 1, -1, 0, -1, -1, 0, -1, 0, 1, 1, 1, 0, 0, 1, 0, -1, -1, 1, 1, 0, 1, -1, 0, 1, -1, 0, 0, -1
central: 0, 0, 0, 1, -1, -1, 1, 0, 0, 0, 0, 1, 0, -1, 1, 1, 0, 1, 1, 0, -1, 1, -1, 1, 0, 0, -1, 0, 0, 1, -1, -1, -1, -1, 1, 1, 1
upper: 0, 0, 0, 1, -1, -1, 1, 0, 0, 0, -1, -1, 1, 0, 0, -1, -1, -1, 1, 0, 0, 1, 0, 0, 1, 1, -1, 0, -1, 0, 0, 0, 0, 1, 1, -1

8. 2,5-Dimethylhexane

lower: 0, 0, 0, 1, -1, -1, 1, 1, -1, 1, 0, 1, -1, 0, 0, -1, 1, -1, -1, -1, 0, 1, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, -1, -1, 1, -1
central: 0, 0, 0, 1, -1, -1, 1, 1, -1, 1, -1, -1, -1, 0, -1, 1, -1, 0, -1, 0, 1, -1, 1, 1, 1, -1, 0, -1, -1, 1, -1, -1, -1, 0, 1, 1
upper: 0, 0, 0, 1, -1, -1, 1, 1, -1, 0, 0, 0, 1, -1, 0, -1, -1, -1, 1, -1, 1, -1, 1, 1, 0, -1, 0, -1, 1, 1, 0, 0, 0, 1, -1, 1, 1, 1

9. 3,3-Dimethylhexane

lower: 0, 0, 0, 1, -1, -1, 1, 1, 1, 0, 1, -1, 1, -1, 1, -1, 1, 1, 0, 0, -1, 1, -1, 1, 1, 0, -1, 1, 0, 1, -1, 0, 0, 1, -1, 1
central: 0, 0, 0, 1, -1, -1, 1, 1, 1, 0, -1, 0, 1, 1, -1, 1, 1, 0, -1, 0, 1, 0, -1, 1, -1, 1, 0, -1, 0, -1, 1, 1, -1, 0, -1, 0, -1
upper: 0, 0, 0, 1, -1, -1, 1, 1, 1, 1, -1, 1, -1, -1, 0, -1, -1, 1, 0, 0, -1, 0, 0, -1, 0, 1, 1, 1, -1, -1, 0, 1, 1, -1, 1, 1, 1

10. 3,4-Dimethylhexane

lower: 0, 0, 0, 1, -1, -1, 1, 0, 1, 0, 0, 0, 1, -1, 1, 0, 1, 0, 0, 1, 0, 0, -1, 0, 0, 1, -1, 0, 0, 0, 1, -1, -1, 1, 0, 0, -1
central: 0, 0, 0, 1, -1, -1, 1, 0, 1, 0, -1, -1, -1, -1, 1, 0, -1, 0, 1, -1, 0, 1, -1, -1, 0, 1, -1, 1, 1, 1, -1, 1, 1, -1, 1, 1
upper: 0, 0, 0, 1, -1, -1, 1, 0, 1, -1, 0, 0, 0, 0, -1, -1, 0, 0, 1, 1, 1, 1, 1, 0, -1, 1, 0, -1, 0, -1, 0, 0, 0, -1, 0, -1, 1, 1

11. 3-Ethylhexane

lower: 0, 0, 0, 1, -1, -1, 1, -1, -1, 0, 1, 1, 0, -1, 1, 0, -1, -1, 0, 0, 1, -1, 0, -1, -1, 0, 1, 0, -1, 0, 1, 1, -1, 0, 1, -1, -1
central: 0, 0, 0, 1, -1, -1, 1, -1, -1, 0, 0, -1, 1, 1, 1, 1, 0, 0, -1, 1, -1, 0, -1, 1, 1, 0, 0, 1, -1, 0, 0, 0, 0, -1, 0, -1, -1
upper: 0, 0, 0, 1, -1, -1, 1, -1, -1, -1, 1, 1, 0, 1, 0, -1, 1, -1, -1, 0, 0, -1, 1, 1, 0, 0, -1, 1, 1, 1, 0, -1, -1, 1, -1, 0, -1

12. 2,2,3-Trimethylpentane

lower: 0, 0, 0, 1, -1, 0, -1, 0, 1, 0, 0, 0, -1, 0, -1, 0, 1, 1, 1, 1, 1, 1, 1, -1, 0, 1, -1, 0, -1, 1, 1, 1, -1, -1, 1, -1

central: 0, 0, 0, 1, -1, 0, -1, 0, 1, -1, 1, 0, 1, 1, 1, -1, 1, 1, 0, 0, 1, 0, -1, 1, 1, -1, -1, -1, 0, 1, -1, 0, 0, -1, -1, 0, 1
upper: 0, 0, 0, 1, -1, 0, -1, 0, 1, -1, -1, 1, 1, 0, -1, 0, 1, -1, 1, 1, 0, 1, 0, -1, 0, 1, 1, 0, -1, 0, 1, 1, 1, 1, 0, -1

13. 2,2,4-Trimethylpentane

lower: 0, 0, 0, 1, -1, 0, -1, 0, 1, -1, -1, 0, 1, -1, 1, 0, 1, 1, -1, -1, 0, -1, -1, -1, -1, 1, 1, 1, -1, 0, 1, 1, 1, 0, -1
central: 0, 0, 0, 1, -1, 0, -1, 0, 0, 1, 0, 1, 1, -1, 1, -1, -1, 0, 0, 0, 0, -1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, -1
upper: 0, 0, 0, 1, -1, 0, -1, 0, 0, 1, -1, -1, 0, 1, -1, 1, 0, 0, -1, 0, 0, -1, 1, 1, 0, 1, 1, 0, -1, -1, 0, 0, 1, -1, 0, 1, -1

14. 2,3,3-Trimethylpentane

lower: 0, 0, 0, 1, -1, 0, -1, 0, -1, 0, 1, 1, 0, 1, 1, -1, 0, 0, -1, 0, -1, -1, 0, 1, 0, 1, -1, 0, -1, -1, -1, 0, 0, 1, 0, 1, -1
central: 0, 0, 0, 1, -1, 0, -1, 0, -1, 0, 0, -1, 0, 0, 1, 1, -1, -1, 0, 0, 1, 0, -1, 1, 1, 1, -1, 0, 1, 1, -1, 0, -1, -1, 0, 0, 1, -1
upper: 0, 0, 0, 1, -1, 0, -1, 0, -1, 0, -1, 0, 0, 0, -1, 1, -1, -1, 1, 1, 0, 0, 0, 1, 0, 1, 0, -1, 1, 0, 0, 1, 0, -1, 1, 1

15. 2,3,4-Trimethylpentane

lower: 0, 0, 0, 1, -1, 0, -1, 0, -1, -1, -1, -1, 1, 1, -1, 0, -1, 0, -1, 1, 0, -1, -1, 1, -1, -1, -1, 1, 1, 0, -1, -1, 0, 0, 0, 0, -1
central: 0, 0, 0, 1, -1, 0, -1, -1, 1, 1, 0, 1, 0, 1, 0, -1, 1, 1, 1, 0, 1, 0, 1, -1, 0, 0, 0, 1, 1, -1, 1, 1, 1, 0, 0, 1
upper: 0, 0, 0, 1, -1, 0, -1, -1, 1, 1, -1, -1, 0, 1, -1, -1, -1, -1, 0, -1, 1, 1, -1, 0, -1, -1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, -1

16. 2,2,3,3-Tetramethylbutane

lower: 0, 0, 0, 1, -1, 0, 1, 0, -1, 1, -1, 1, 1, 0, -1, 0, 1, -1, 0, 0, -1, -1, 1, 1, -1, -1, 1, 0, -1, 1, 1, -1, 0, -1, -1
central: 0, 0, 0, 1, -1, 0, 1, 0, -1, 0, 0, 1, 1, -1, -1, 1, 1, 0, -1, 1, -1, 0, 0, -1, 1, 0, -1, -1, -1, 1, 0, 0, -1, -1, 0, -1, 0, 1
upper: 0, 0, 0, 1, -1, 0, 1, 0, -1, -1, 1, 1, 0, 1, 0, 0, 1, 1, 1, -1, 1, 0, 0, 1, -1, 0, -1, 0, 0, 1, 0, 1, -1, 0, 1, 0, -1

17. 3-Ethyl-2-methylpentane

lower: 0, 0, 0, 1, -1, -1, 1, 1, -1, -1, 1, 0, -1, 0, -1, -1, 0, -1, -1, 0, 1, -1, 1, 0, 0, 1, 1, 0, 0, -1, 1, 1, 1, -1
central: 0, 0, 0, 1, -1, -1, 1, 1, -1, -1, -1, 1, 0, -1, 0, 1, 1, 1, -1, 1, -1, 0, -1, 0, -1, 0, 1, 0, 0, -1, -1, -1, 1, 1, 0, -1
upper: 0, 0, 0, 1, -1, -1, 1, 0, 1, 1, 1, 0, 0, 0, -1, 0, -1, -1, 0, -1, -1, 1, 0, 1, 1, 0, 0, 0, 0, -1, 0, -1, 0, -1, 1, 1, 1

18. 3-Ethyl-3-methylpentane

lower: 0, 0, 0, 1, -1, 0, -1, 0, 0, 1, -1, 1, -1, 0, -1, 0, 1, -1, 1, 0, 0, 1, 0, -1, 1, 0, 1, -1, -1, 1, -1
central: 0, 0, 0, 1, -1, 0, -1, 0, 0, 0, 1, -1, -1, -1, -1, -1, 0, 0, 0, -1, -1, 0, 0, 0, -1, 0, 0, 1, -1, 0, 0, 0, 1, 1, -1
upper: 0, 0, 0, 1, -1, 0, -1, 0, 0, 0, -1, -1, 1, 0, 1, 0, 0, 1, 0, 1, 0, 0, -1, 0, 1, 1, 1, 0, 1, 0, 1, -1, -1, 0, 1, 1, -1